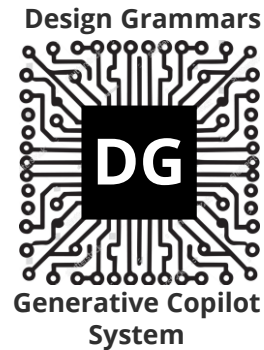


ptBIM 26

MANAGEMENT OF THE DESIGN KNOWLEDGE CORE IN A CROSS-PLATFORM ENVIRONMENT OF THE BIM PROJECT



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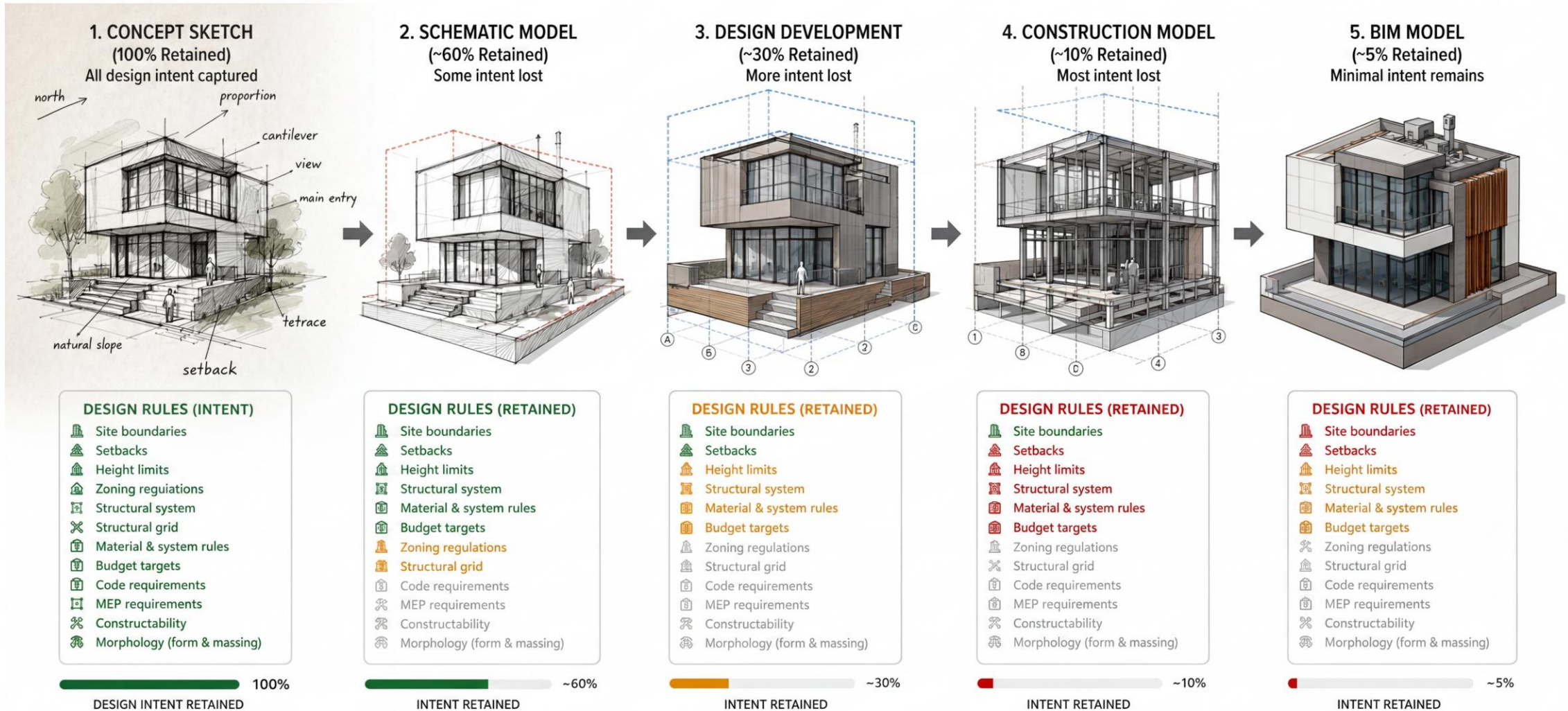
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- 1. The problem — fragmented design knowledge**
- 2. Proposed framework — Shared Knowledge Core**
- 3. System architecture — five layers, two domains**
- 4. Case study — plan-layout generation**
- 5. Conclusion & discussion**

How much design intent survives the next handover?
We exchange files. The rules behind them do not travel.

How much design intent survives the next handover?
We exchange files. The rules behind them do not travel.

FROM SKETCH TO BIM – STRIPPING AWAY DESIGN RULES

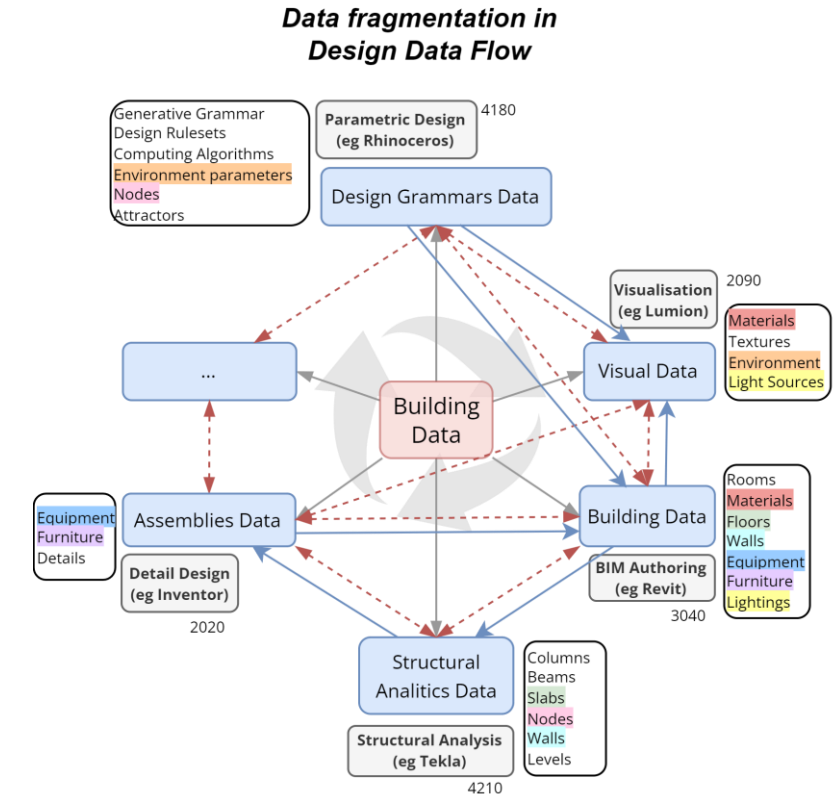


LEGEND:

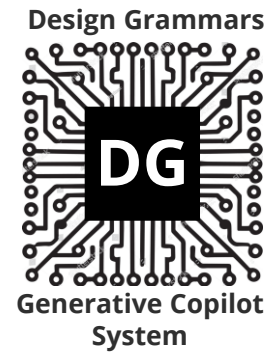
-  Retained
Design rule carried forward
  Partially retained
Design rule weakened or not fully enforced
  Lost
Design rule not carried forward
  Stripped
Design rule removed or no longer tracked

THE PROBLEM: DESIGN KNOWLEDGE GETS LOST IN TRANSLATION

- BIM automates information — but design knowledge stays fragmented across files, platforms and disciplines.
- Each handover transfers only the final state, not the rules and constraints that produced it.
- IFC handles physical components well, but parametric relationships and abstract constraints fall outside its schema.
- Parametric tools (Grasshopper / Dynamo) replicate logic per platform rather than sharing it.



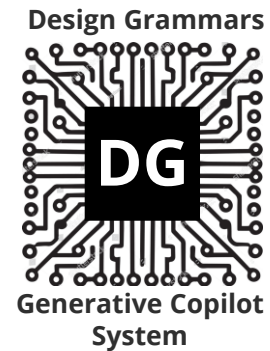
FROM FILE-BASED EXCHANGE TO OBJECT-LEVEL KNOWLEDGE SHARING



Goal

Build a Shared Knowledge Core that carries design intent across tools and phases — not just geometry.

FROM FILE-BASED EXCHANGE TO OBJECT-LEVEL KNOWLEDGE SHARING

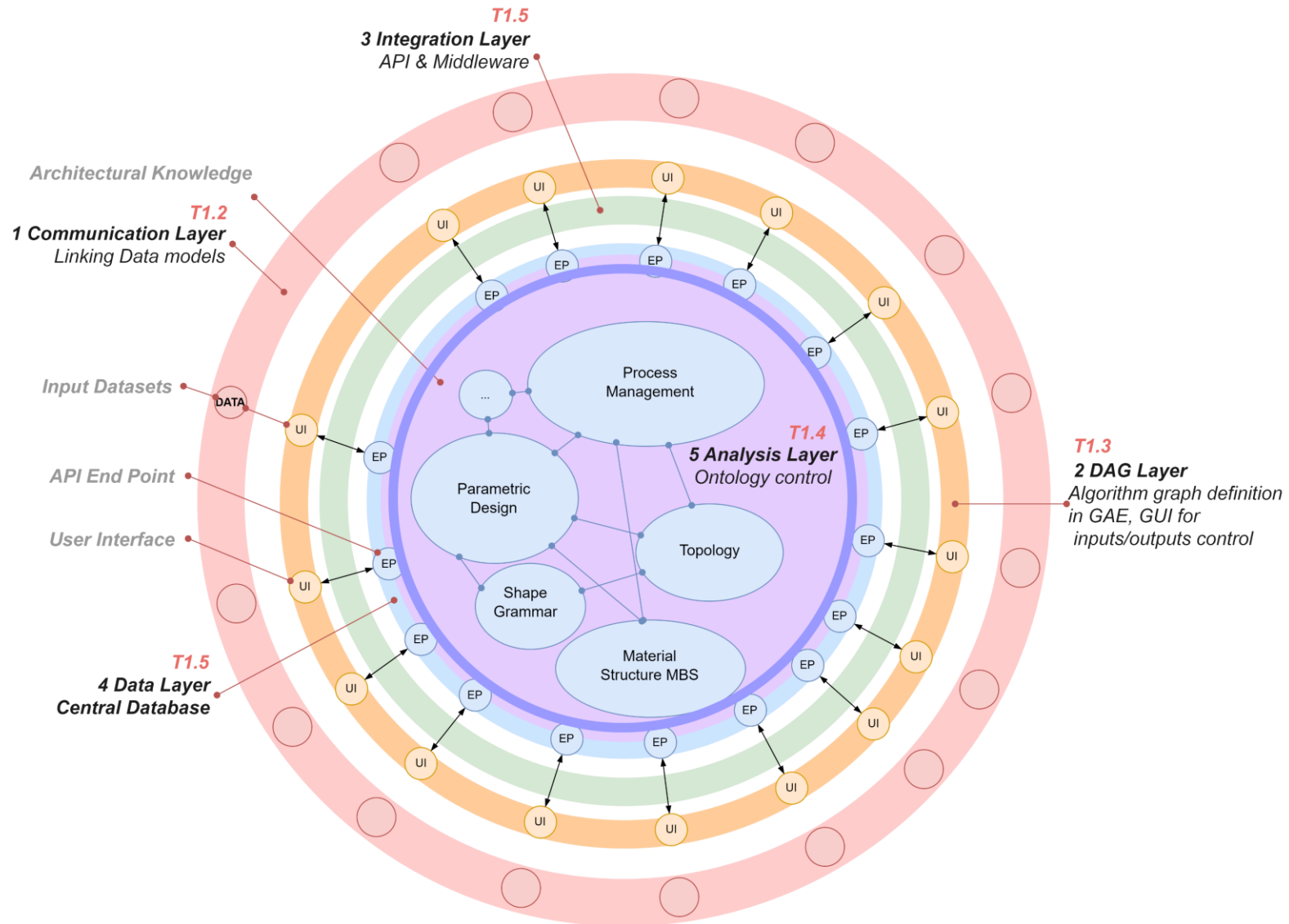


Approach

- Core-Periphery data model
- Abstract design objects + rules at the centre
- Domain-specific representations (BIM, parametric, analysis) at the edge
- Object-level semantic communication replaces file exchange

SYSTEM FRAMEWORK — FIVE LAYERS

Replaces file-based exchange with object-level semantic communication.



TWO INTERFACES, ONE KNOWLEDGE CORE

Model Domain

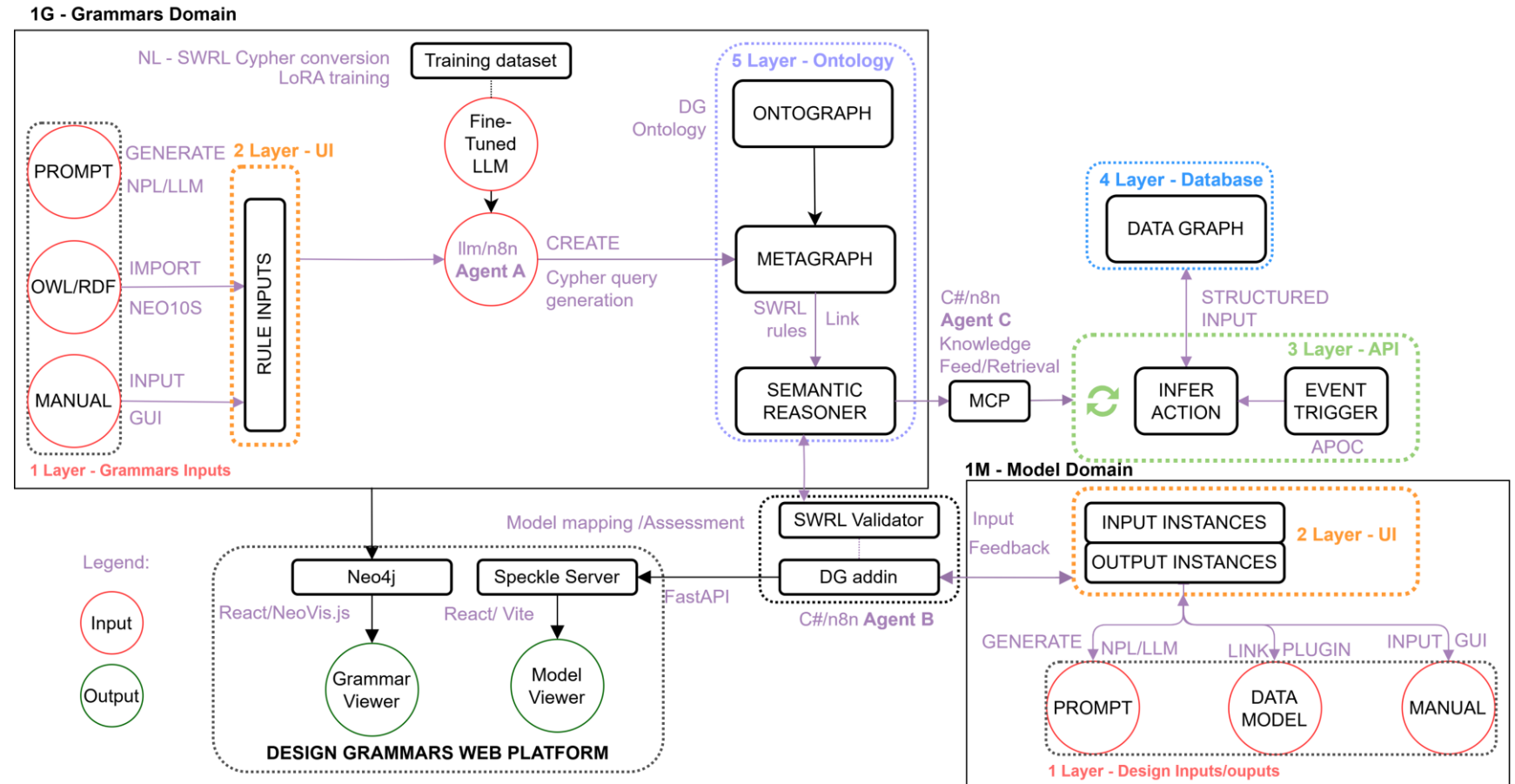
- BIM geometry & properties
- IFC-aligned class hierarchy

Grammar Domain

- Design rules in natural language → SWRL
- Parsed by constrained LLM
- Validated against the ontology

Bridge

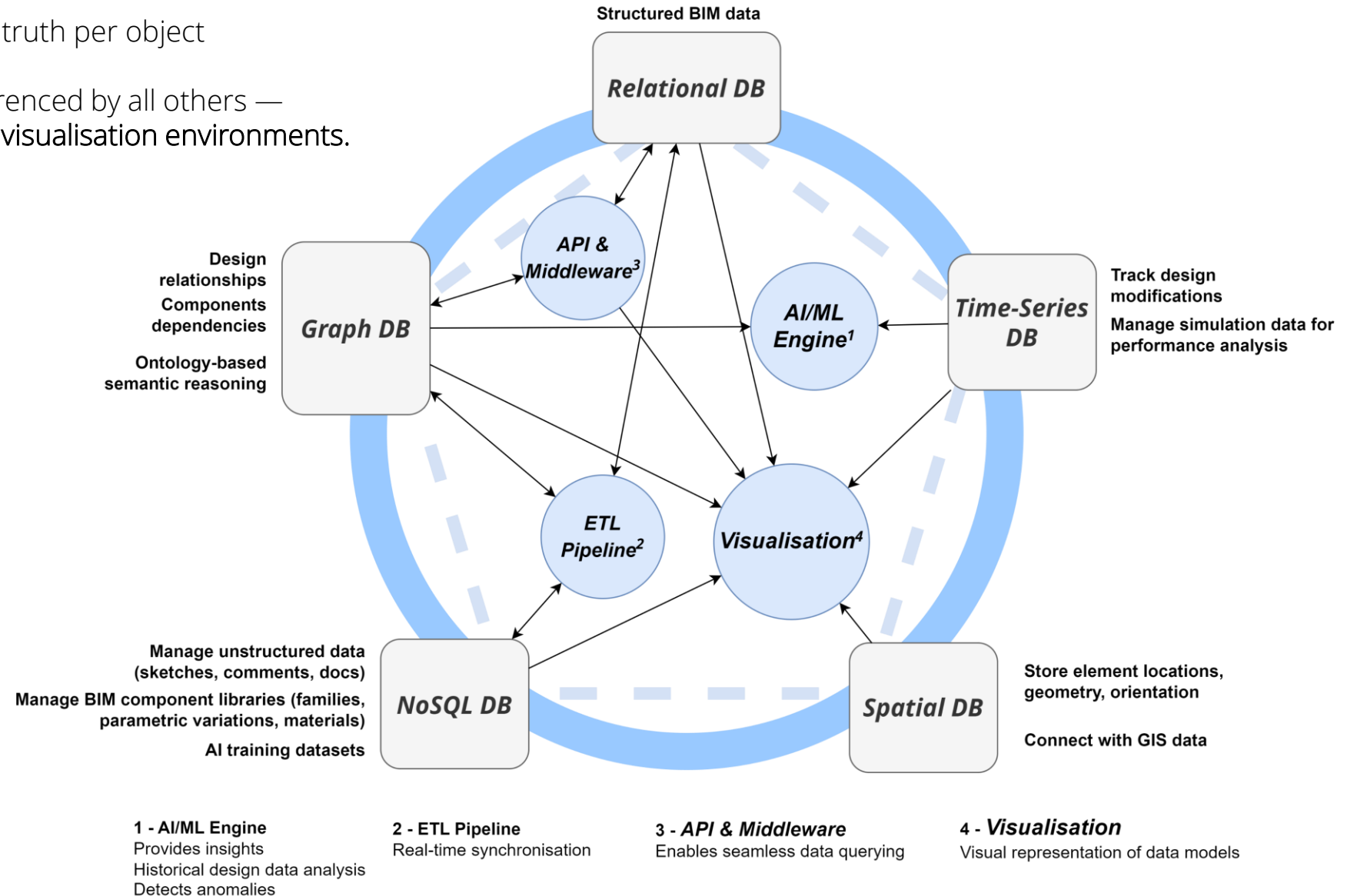
- n8n workflow + MCP protocol
- Live Grasshopper / Rhino.Inside connector



DATA LAYER — SINGLE-SOURCE OBJECT INTEGRITY

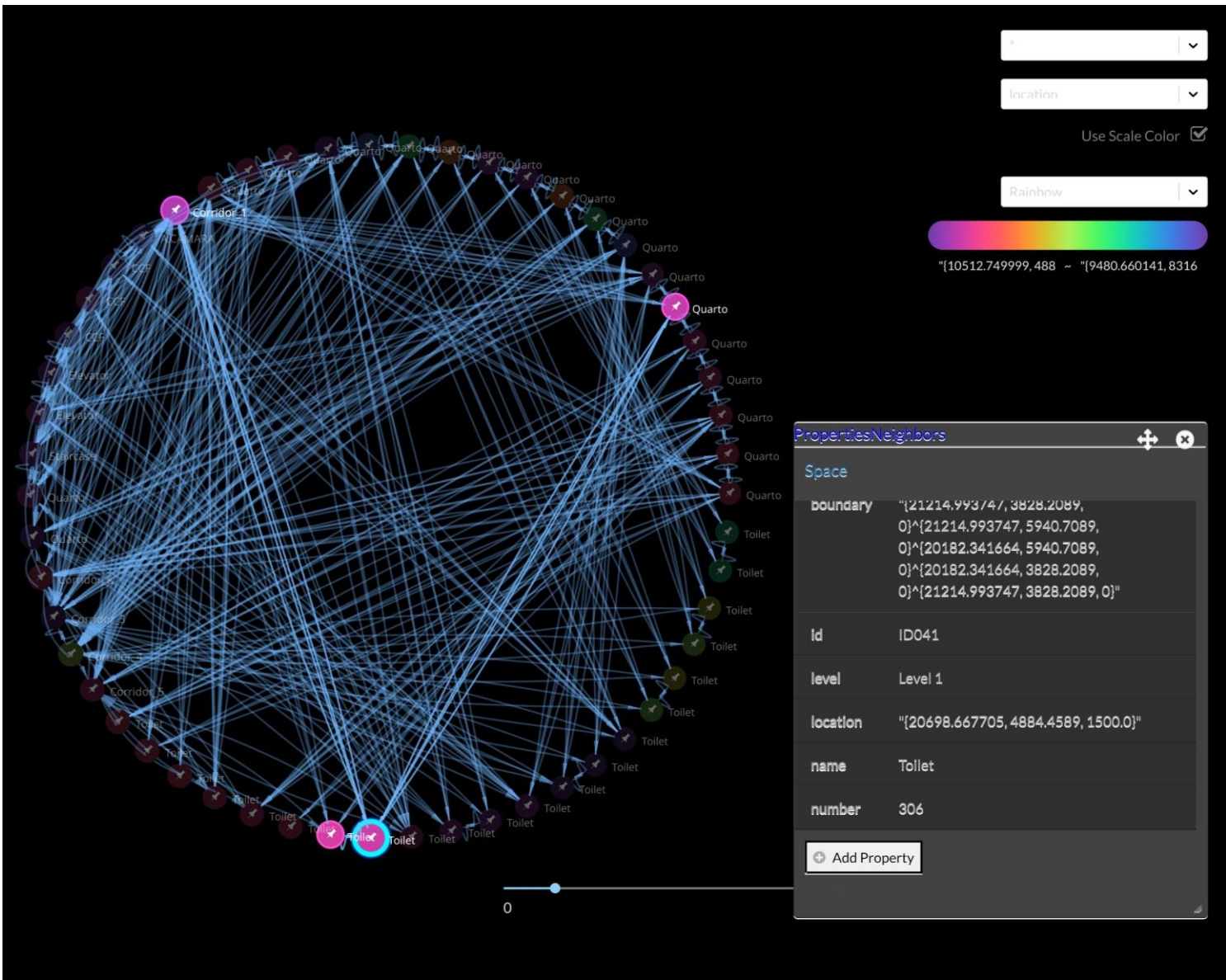
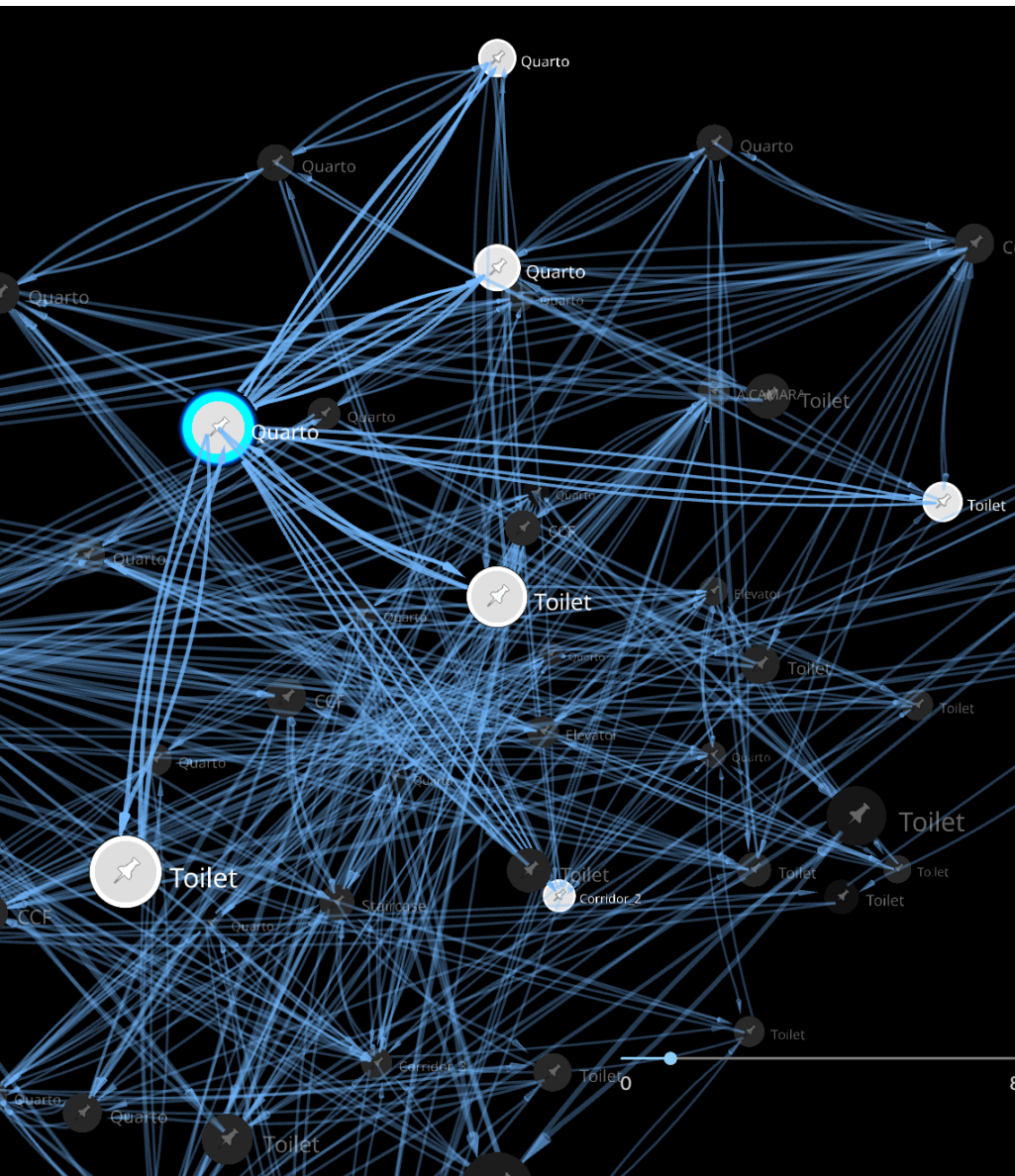
Multi-domain storage, one source of truth per object

Each object is defined once and referenced by all others —
no replication across BIM / analysis / visualisation environments.



USE CASE 1 RESIDENTIAL PLAN-LAYOUT GENERATION

Spatial program encoded once → reused to generate a solution space of valid alternatives, dynamically evaluated as parameters change.

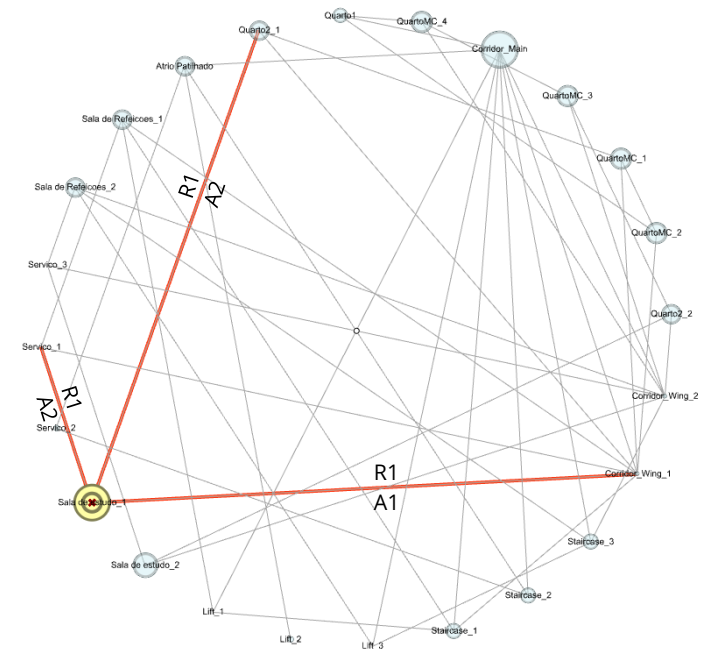


USE CASE 1 RESIDENTIAL PLAN-LAYOUT GENERATION

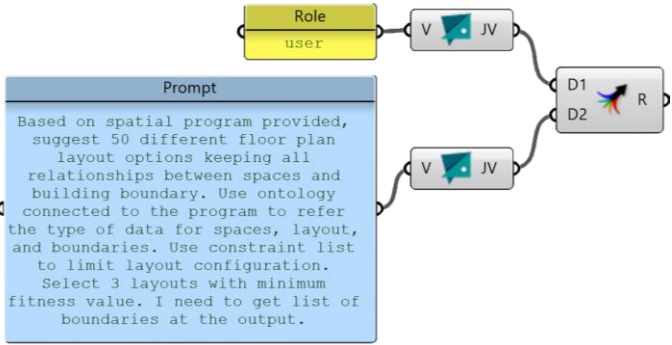
Spatial program encoded once → reused to generate a solution space of valid alternatives, dynamically evaluated as parameters change.

Extracted from Graph Room Schedule

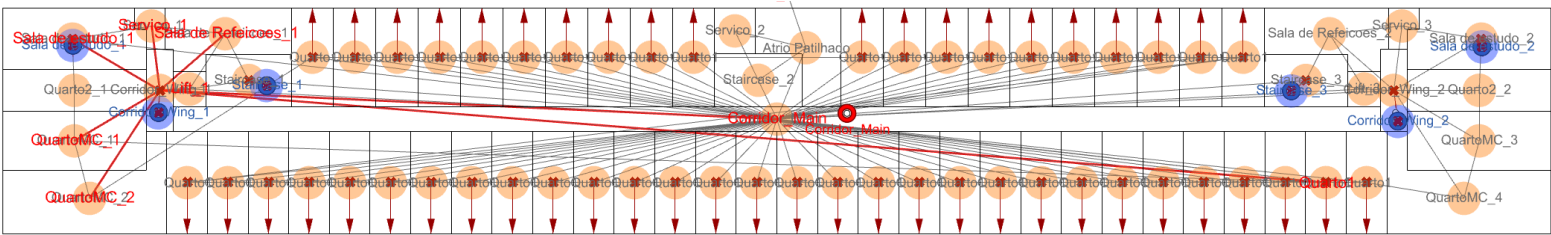
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1	ID	Name	Area	Location	Boundary	Level	Number
2	ID001	Quarto	12.2491	2, 8258.215, 0}^{272	Level 1		311
3	ID002	Quarto	12.2464	4, 4942.16771, 0}^{29	Level 1		310
4	ID003	Quarto	13.3971	6, 1841.508, 0}^{272	Level 1		309
5	ID004	Quarto	14.0623	1, 2274.8211664, 382	Level 1		308
6	ID005	Quarto	14.0636	1, 2274.9693747, 594	Level 1		307
7	ID006	Quarto	13.3845	4, 2257.182396, 5940	Level 1		306
8	ID007	Quarto	20.3524	9, 1787.35, 2511.208	Level 1		305
9	ID008	Quarto	14.0561	3, 2273.570, 0}^{7549	Level 1		303
10	ID009	Quarto	14.126	2, 2266.605039.9723	Level 1		302
11	ID010	Quarto	11.9355	1, 2777.468, 225.0, 0}	Level 1		301
12	ID011	Quarto	13.7273	3, 3047.793, 6600.052	Level 1		300
13	ID012	Quarto	13.7276	, 10152.8293747, 66	Level 1		320
14	ID013	Quarto	11.9393	10422.680, 8020.70	Level 1		319
15	ID014	Quarto	14.119	, 10935.327549.9786	Level 1		318
16	ID015	Quarto	14.0478	, 10928.1205258, 0}^{	Level 1		317
17	ID016	Quarto	16.8605	1, 12391.66, 10575.7	Level 1		316
18	ID017	Quarto	14.0931	2, 10919.747, 13890	Level 1		314
19	ID018	Quarto	14.0669	5, 10925.06993747, 7	Level 1		313
20	ID019	Quarto	13.4057	2, 11359.678, 0}^{261	Level 1		312
21	ID020	Corridor_1	10.3566	7, 6624.749, 0}^{188	Level 1		326
22	ID021	A.CAMARA	3.24784	9, 9860.7278618, 10	Level 1		322
23	ID022	CCF	3.78915	6, 8170.7117, 0}^{99	Level 1		323
24	ID023	CCF	3.60921	2, 8170.7189, 0}^{191	Level 1		325
25	ID024	CCF	3.54496	8, 4774.6018.923958	Level 1		327
26	ID025	Elevator	4.05436	17, 4805.897089, 0}^{	Level 1		328
27	ID026	Elevator	4.37193	33, 4805.8917451.4	Level 1		329
28	ID027	Staircase	23.2376	5, 8950.71, 10575.72	Level 1		321
29	ID028	Quarto	16.7191	7, 11873.99, 12773.2	Level 1		315
30	ID029	Quarto	21.4167	9, 1916.111.2089, 0}	Level 1		304
31	ID030	Corridor_2	2.49341	9, 6600.7457119, 0}^{	Level 1		326
32	ID031	Corridor_3	8.30438	5, 6609.487958, 5940	Level 1		326
33	ID032	Corridor_4	6.21868	3, 6600.69458, 0}^{26	Level 1		326



Spatial relationships typology graph (Grasshopper)



Prompt query to the system (Grasshopper)



Legend: → Functional space attractors ● Geometrical space attractors ● Entrance, End points Relationships Boundary

USE CASE 2 DESIGN STATE VALIDATION

USE CASE — FROM CLIENT REQUIREMENTS TO VALIDATED DESIGN

Three stakeholders · four acts · Knowledge Graph → MetaGraph → ValidationGraph

1 CLIENT BRIEF

Requirements & constraints uploaded to DG

2 RULE AUTHORIZING

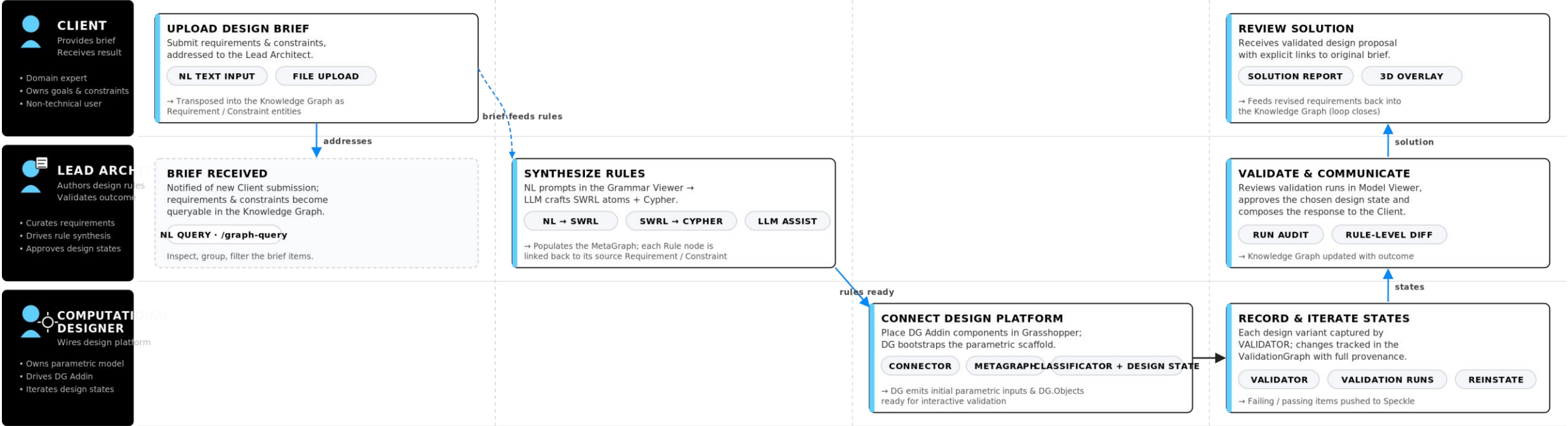
NL prompts → SWRL → Cypher → MetaGraph

3 DESIGN BOOTSTRAP

Parametric inputs + DG objects generated

4 VALIDATE & COMMUNICATE

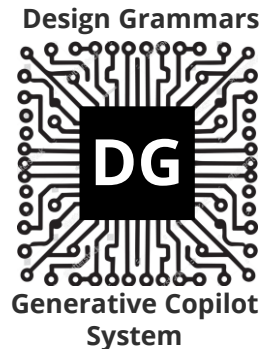
Design states tracked · solution returned



CONCLUSION

Contribution

A working prototype that encodes, manages and reuses grammar knowledge across the design lifecycle.



What works

- Plan-layout generation from a reusable spatial program
- Iterative validation of design states against rule sets
- Structured ingestion of project documentation

Limitations & next steps

- SWRL coverage limited to ClassAtom / DataPropertyAtom / BuiltinAtom
- LLM Cypher output still needs human review
- Academic evaluation only — production stress-testing pending
- IDS integration left for future work

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